Project Proposal Requirements

Final Output in Project Management Information Systems

Development of a Web-Based Multi-Branch Store Management Information System for IGLÚ for Sales and Employee Monitoring in Barangay Salapungan, Angeles City

Members:

Souguilon, Carl Asnef B.

Fajardo, Roanne E.

Garcia, Austin Karl C.

Ignacio, Kariel Justine C.

Recapor, John Paul A.

Prepared For:

Jan Catherine S. Santiago

November 3, 2026



INSTITUTE OF COMPUTING STUDIES AND LIBRARY INFORMATION SCIENCE

Table of Contents

Project Title **3**
Background of the Project and Problem Statement **3**
A. Background of the Study 3
B. Statement of the Problem 4
C. Specific Problems 4
Project Objectives **5**
General Objective 5
Specific Objectives 5
Target Users or Beneficiaries **6**
Scope and Limitations **8**
A. Scope 8
B. Limitations 9
Expected Project Output **10**
A. Administrator and Management Features 10
B. Employee and Cashier Features 11
C. System and Data Features 12
Project Team and Responsibilities **12**
Project Activities and Work Breakdown Structure **14**
Gantt Chart **15**
Resources and Estimated Budget **16**
A. Required Resources 17
B. Preliminary Estimated Project Budget 17
C. Budget Control 19
Possible Risks and Proposed Solutions **19**
Communication Plan **24**
PMIS Tool to Be Used **25**
Expected Benefits **29**
Conclusion and References **30**
Conclusion 30
References 32
Appendices **33**

Project Title

Development of a Web-Based Multi-Branch Store Management Information System for IGLÚ for Sales and Employee Monitoring in Barangay Salapungan, Angeles City

Background of the Project and Problem Statement

A. Background of the Study

Businesses operating across multiple branches need an organized system to manage employee information, attendance, sales, and daily operations. As businesses expand, maintaining accurate and consistent records across different locations becomes increasingly important. Effective information management helps ensure that business records are accessible, reliable, and easier to monitor.

In multi-branch store operations, managing employee records, attendance, and cashier transactions separately may lead to data inconsistencies, delays in updating information, and difficulties in monitoring branch performance. Manual recordkeeping and disconnected digital files may also limit the accessibility of information and increase the risk of unauthorized modifications to important records. These challenges highlight the need for a centralized Management Information System (MIS) that can organize records and support monitoring across multiple branches.

IGLÚ, a self-service frozen yogurt and ice cream business, requires the management of sales, employees, attendance, and cashier-related operations across its branches. When these records are maintained separately, monitoring business activities and ensuring consistency among records may become challenging. These circumstances establish the need to examine how IGLÚ

manages its operational information and identify opportunities to improve its existing processes.

To address these concerns, the proponents propose **IgluHub**, a Web-Based Multi-Branch Store Management Information System designed to centralize employee records, attendance monitoring, cashier-related records, and branch information. The proposed system aims to improve information organization, support consistent recordkeeping, and provide authorized users with a more efficient way to monitor store operations in Barangay Salapungan, Angeles City.

B. Statement of the Problem

The project aims to address the difficulties associated with managing employee records, attendance, cashier-related information, and branch operations when these records are maintained separately. Such practices may contribute to delays, inconsistent information, and difficulties in monitoring business activities. They may also expose important sales and attendance records to unauthorized modifications.

Specifically, the project seeks to answer the following questions:

C. Specific Problems

1. **Employee Records Management**
   How can employee information be organized and managed efficiently across different branches?
2. **Attendance Monitoring**
   How can employee attendance be recorded and monitored consistently across branches?

3. **Cashier Operations Monitoring**
   How can cashier-related records and sales transactions be organized and monitored more efficiently?
4. **Branch Monitoring**
   How can authorized users access and monitor branch information through a centralized system?
5. **Data Consistency**
   How can the consistency and timely updating of records across different branches be improved?
6. **Unauthorized Data Modification**
   How can the system help prevent and detect unauthorized changes to sales and attendance records?

Project Objectives

General Objective

To develop IgluHub, an integrated multi-branch Store Management Information System that provides a centralized, organized, and secure platform for managing employee records, attendance, and cashier operations across different branches.

Specific Objectives

The proposed project specifically aims to:

1.To develop a centralized employee record management feature that allows authorized personnel to efficiently add, update, view, and manage employee information across different branches.

2. To develop an attendance management feature that allows authorized personnel to record, monitor, and review employee attendance records, including time-in, time-out, absences, and tardiness.

3. To develop a cashier management feature that allows authorized personnel to record, organize, and monitor cashier-related transactions and sales records.

4. To provide a centralized platform that allows authorized management personnel to access and monitor employee, attendance, and cashier information from different branches in one system.

5.To implement centralized data storage and validation mechanisms to minimize inconsistent, duplicated, delayed, or missing information between branches.

6.To implement user access controls and audit trail mechanisms that restrict unauthorized changes and record user activities involving sales and attendance data, thereby improving accountability and facilitating the review of suspicious or unauthorized modifications.

Target Users or Beneficiaries

The proposed **Development of a Web-Based Multi-Branch Store Management Information System for IGLÚ for Sales and Employee Monitoring in Barangay Salapungan, Angeles City** aims to improve the manual tracking of employee information, attendance, and transactions across Iglu branches. The system is designed to provide centralized and organized records that can support branch operations, improve information accessibility, and assist management in monitoring business activities. The following are the identified target users and beneficiaries of the proposed system:

Iglu Branch Owners – They will benefit from having a centralized view of branch operations, allowing them to monitor business activities and make informed decisions based on organized records.

Head Office Management – The system will help management consolidate information from different branches, making it easier to review operational records and compare branch performance.

Iglu Branch Employees – Employees will benefit from a more organized system for maintaining their attendance records and relevant work-related information.

Iglu Customers – Customers may benefit from more organized transaction processing and improved service efficiency as the system helps streamline cashier operations.

Business Operations Management – The system may support better coordination among branches by providing centralized access to employee, attendance, and transaction information.

Business Owners Planning to Expand – Business owners who plan to establish additional branches may use the system as a model for managing records and monitoring operations across multiple locations.

Information Systems Practitioners – IT professionals and system developers may use the project as a reference for designing information systems that address manual recordkeeping and multi-branch monitoring.

Small and Medium-Sized Retail Businesses – Similar businesses that rely on manual employee records, attendance tracking, and transaction documentation may gain insights into the benefits of centralized information management.

Academic Institutions and Instructors – The study may serve as a learning reference for students and instructors discussing systems analysis, information systems development, database management, and business process improvement.

Future Researchers – Future researchers may use the study as a reference for developing or enhancing systems related to multi-branch store management, employee monitoring, attendance tracking, and transaction management. It may also serve as a foundation for further studies on centralized information systems and the improvement of manual business processes

Scope and Limitations

A. Scope

The project covers the design, development, and testing of IgluHub, a web-based Store Management Information System for participating IGLÚ branches in Barangay Salapungan, Angeles City. The system will use a centralized database and role-based access to support the following functions:

- **Employee records:** create, view, update, search, assign to a branch, activate, or deactivate employee accounts according to permissions.
- Attendance: record time-in and time-out, view attendance history, and review recorded absences and tardiness.
- **Cashier transactions:** enter multiple items in a sale; record relevant product details, weight or quantity, prices, discounts, payment method, and GCash reference information when applicable; and calculate totals and cash change.
- **Transaction review and correction:** allow authorized personnel to review transactions and require manager authorization for sensitive edits or voids. Corrections and voids will be logged rather than silently overwriting the original record.

- **Branch management:** maintain branch information, assign employees to branches, and filter permitted reports by branch and date.
- **Dashboard and reports:** present relevant summaries of sales, transaction counts, attendance, payment methods, and product performance based on the available and verified records.
- **Security and accountability:** apply role-based permissions, input validation, session controls, and audit logging for important actions.

B. Limitations

The system requires an internet connection for normal access and synchronization with the centralized database. During an outage, updates may be delayed or unavailable; offline transaction processing is not included unless separately approved and developed.

- System accuracy depends on correct data entry, verified product prices, correct employee assignments, and agreed business rules supplied by IGLÚ.
- The system will support recording payment details, including cash and GCash references where applicable; it will not directly connect to or authorize payments through a bank or e-wallet unless a separate approved integration is added.
- The project will not include payroll computation, inventory purchasing, supplier management, accounting, customer loyalty accounts, or automated tax filing unless these are later approved as part of the project scope.
- The system's security depends on appropriate configuration, secure account practices, database permissions, and testing. It can reduce risks but cannot guarantee that all unauthorized activity or data loss will be prevented.
- Performance and availability depend on the selected hosting and database services, the devices used by the branches, and the quality of their internet connections.
- Pilot testing and evaluation will be limited to the branches, users, devices, and period approved by IGLÚ and the project adviser.

Expected Project Output

The expected output is IgluHub, a responsive web-based multi-branch Store Management Information System. The features below define the proposed deliverables and are aligned with the objectives and scope.

A. Administrator and Management Features

1. Secure Login and Role-Based Access – Restrict functions and records according to user roles and permissions.
2. Management Dashboard – Summarize relevant sales, transaction counts, attendance status, and branch activity.
3. Branch Management – Add, view, update, and manage participating branch records.
4. Employee Management – Create, update, search, activate, and deactivate employee accounts.
5. Branch Assignment – Assign employees to the appropriate branch and manage assignments according to permissions.
6. Attendance Monitoring – View attendance history, time-in/time-out records, absences, and tardiness by employee, date, or branch.
7. Sales Monitoring – Search and review recorded transactions and view relevant transaction details.
8. Branch and Date Filters – Narrow permitted dashboard information and reports by branch and date range.
9. Reports and Analytics – Present summaries of revenue, transaction counts, attendance, payment methods, and product performance using recorded data.
10. Audit Log Review – Allow authorized personnel to review logged sensitive actions, including transaction edits, voids, and changes to important records.

B. Employee and Cashier Features

1. Employee Login – Provide secure, individual access to permitted employee functions.
2. Attendance Time-In and Time-Out – Record an employee's attendance events and display the resulting history.
3. Cashier Transaction Entry – Record a customer sale with multiple items in a single transaction.
4. Product and Pricing Details – Record the relevant product type, weight or quantity, price, and applicable size or item details according to IGLÚ's confirmed pricing rules.
5. Automatic Total Calculation – Calculate line totals and transaction totals based on the entered items and configured prices.
6. Discount Handling – Apply eligible Senior Citizen or PWD discounts according to the business rules confirmed by IGLÚ and applicable requirements.
7. Payment Recording – Record cash or GCash as the payment method and capture a GCash reference number when applicable; the system will record payment information rather than directly process the payment.
8. Change Calculation – Calculate the change due for cash transactions when the amount tendered exceeds the total due.
9. Transaction Review – Display the transaction details and total for cashier verification before submission.
10. Manager Authorization – Require an authorized manager to approve sensitive transaction edits or voids.
11. Transaction Correction and Void – Preserve accountability by recording authorized corrections or voids and their reasons in the audit log.
12. Attendance History – Allow employees to view their own attendance records, subject to system permissions.

C. System and Data Features

1. Centralized Database – Store employee, attendance, branch, product/pricing, and transaction records in a shared database.
2. Data Validation – Check required fields, valid formats, and reasonable values before saving records.
3. Branch Association – Associate employees, attendance events, and transactions with the appropriate branch.
4. Permission Enforcement – Restrict sensitive functions and records to authorized users.
5. Audit Logging – Record the user, action, date and time, affected record, and relevant correction details for defined sensitive actions.
6. Data Synchronization – Make saved information available to authorized users through the centralized service, subject to connectivity and service availability.
7. Responsive Interface – Support use through common desktop and mobile web browsers, subject to device and browser compatibility.
8. Backup and Recovery Planning – Use the backup capabilities available from the selected hosting/database provider and document a basic recovery procedure.

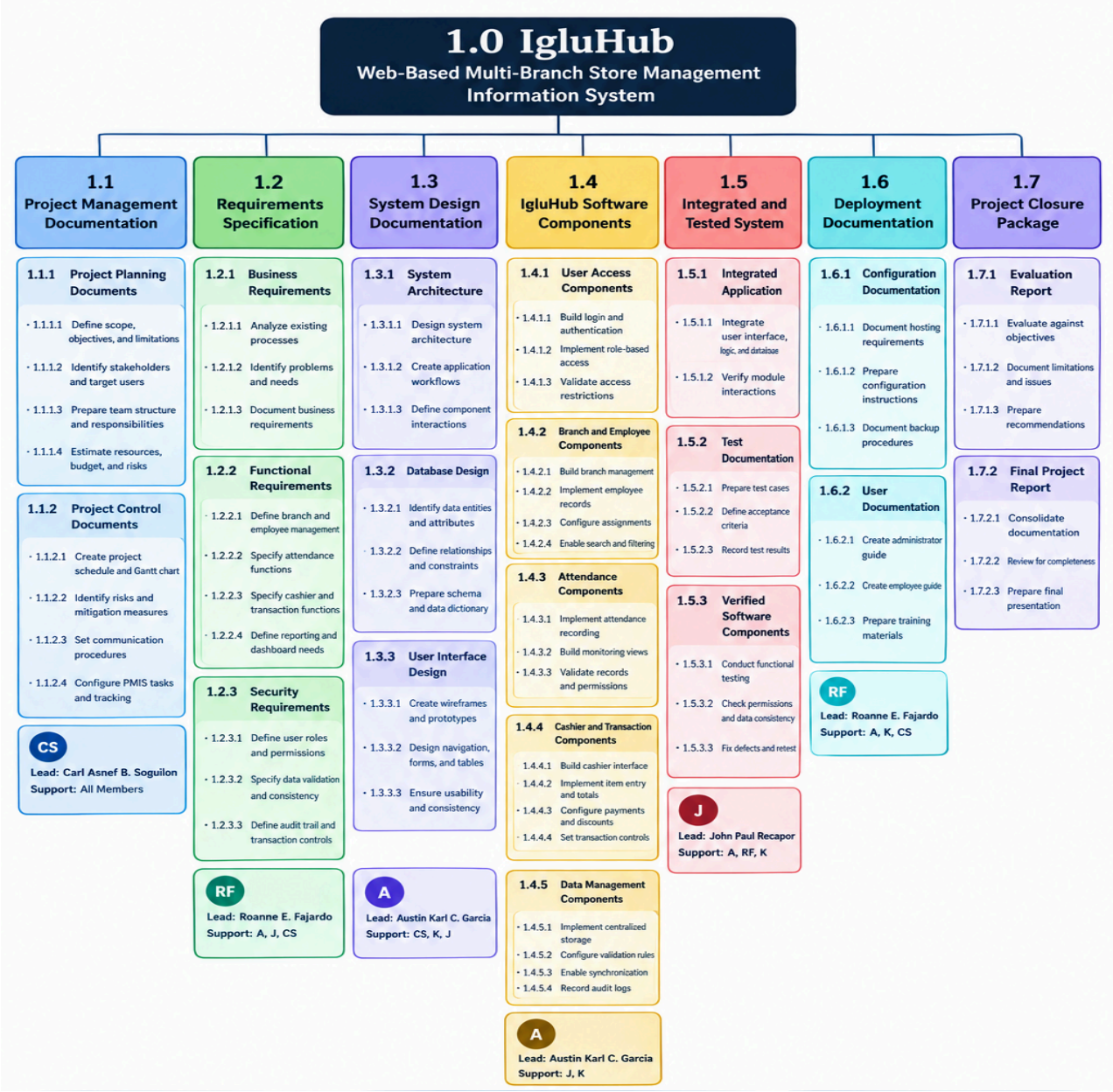
Project Team and Responsibilities

The five-member team will collaborate on requirements analysis, proposal preparation, system planning, development, testing, and documentation. Each member has a primary responsibility, while decisions affecting the system's requirements and scope will be reviewed by the whole team.

Team Member	Designated Role	Responsibilities
Carl Asnef B. Soguilon	Project Manager and Systems Analyst	Coordinates the team, organizes meetings and task assignments, maintains the overall project direction, leads requirements analysis, checks consistency

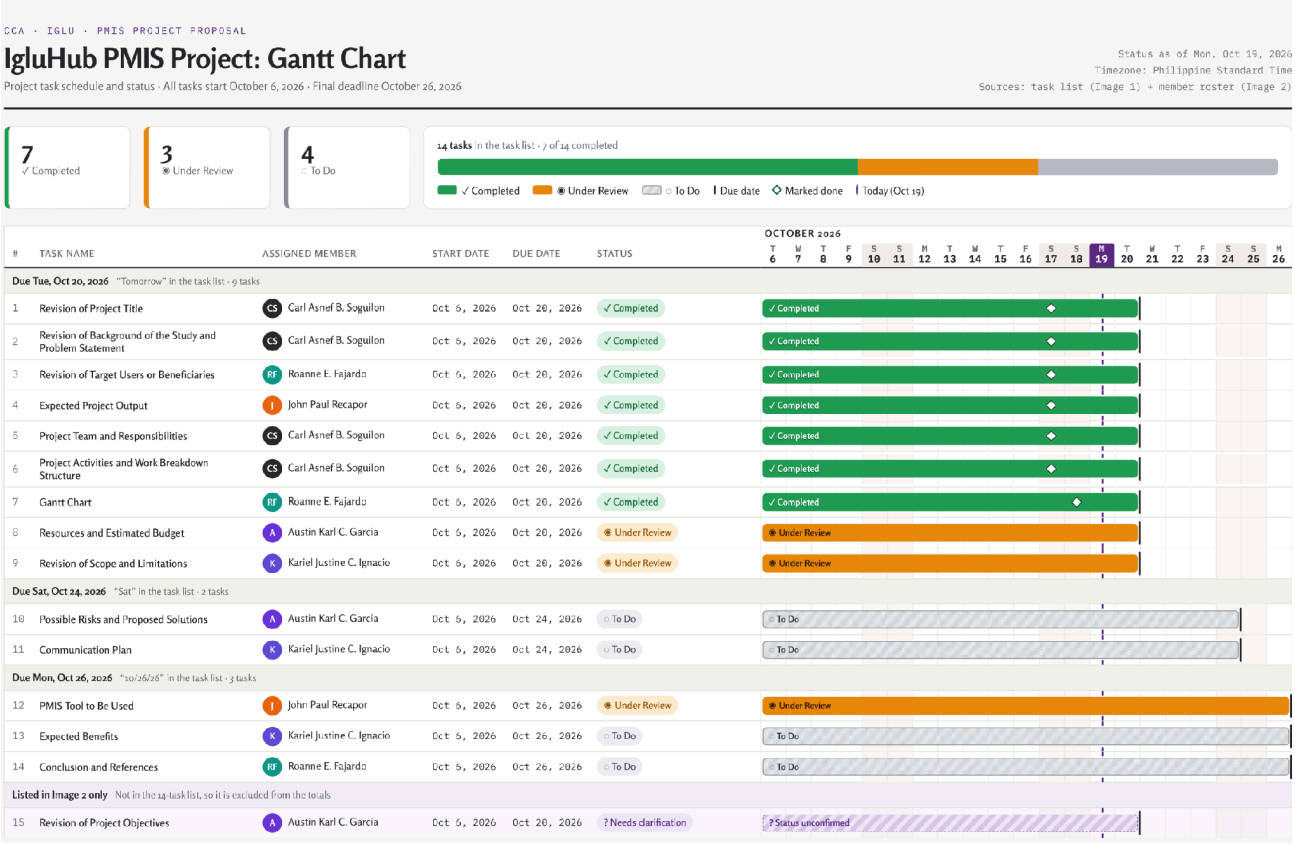
		among proposal sections, and monitors completion of agreed deliverables.
Roanne E. Fajardo	Documentation and Schedule Coordinator	Maintains organized project documentation, consolidates revisions, records decisions and meeting notes, and coordinates the schedule documentation with the team.
Austin Karl C. Garcia	Technical Planning and Risk Coordinator	Reviews technical feasibility, documents system and database requirements, identifies project risks and mitigations, and prepares the resource and budget estimates with team input.
Kariel Justine C. Ignacio	Scope and Communication Coordinator	Maintains the scope and limitations, coordinates communication with the adviser and client representatives, documents requirements clarifications, and helps ensure that proposed features remain within the agreed boundaries.
John Paul Recapor	System Output and PMIS Research Coordinator	Documents the expected system outputs, evaluates and documents ClickUp as the PMIS tool, and supports task tracking, progress reporting, and coordination of project deliverables.

Project Activities and Work Breakdown Structure



Gantt Chart

Figure A — Proposal Documentation Schedule (actual ClickUp workspace screenshot, tracking the writing and revision of this proposal document; not the system-implementation timeline).



CCA · IGLU · PMIS PROJECT PROPOSAL

IgluHub PMIS Project: Gantt Chart

Project task schedule and status · All tasks start October 6, 2026 · Final deadline October 26, 2026

Status as of Mon, Oct 19, 2026
Timezone: Philippine Standard Time
Sources: task list (Image 1) + member roster (Image 2)

7	3	4
✓ Completed	◉ Under Review	○ To Do

14 tasks in the task list · 7 of 14 completed

✓ Completed · ◉ Under Review · ○ To Do · Due date · ◇ Marked done · Today (Oct 19)

#	TASK NAME	ASSIGNED MEMBER	START DATE	DUE DATE	STATUS
	Due Tue, Oct 20, 2026 "Tomorrow" in the task list · 9 tasks				
1	Revision of Project Title	Carl Asnef B. Soguilon	Oct 6, 2026	Oct 20, 2026	✓ Completed
2	Revision of Background of the Study and Problem Statement	Carl Asnef B. Soguilon	Oct 6, 2026	Oct 20, 2026	✓ Completed
3	Revision of Target Users or Beneficiaries	Roanne E. Fajardo	Oct 6, 2026	Oct 20, 2026	✓ Completed
4	Expected Project Output	John Paul Recapor	Oct 6, 2026	Oct 20, 2026	✓ Completed
5	Project Team and Responsibilities	Carl Asnef B. Soguilon	Oct 6, 2026	Oct 20, 2026	✓ Completed
6	Project Activities and Work Breakdown Structure	Carl Asnef B. Soguilon	Oct 6, 2026	Oct 20, 2026	✓ Completed
7	Gantt Chart	Roanne E. Fajardo	Oct 6, 2026	Oct 20, 2026	✓ Completed
8	Resources and Estimated Budget	Austin Karl C. Garcia	Oct 6, 2026	Oct 20, 2026	◉ Under Review
9	Revision of Scope and Limitations	Kariel Justine C. Ignacio	Oct 6, 2026	Oct 20, 2026	◉ Under Review
	Due Sat, Oct 24, 2026 "Sat" in the task list · 2 tasks				
10	Possible Risks and Proposed Solutions	Austin Karl C. Garcia	Oct 6, 2026	Oct 24, 2026	○ To Do
11	Communication Plan	Kariel Justine C. Ignacio	Oct 6, 2026	Oct 24, 2026	○ To Do
	Due Mon, Oct 26, 2026 "10/26/26" in the task list · 3 tasks				
12	PMIS Tool to Be Used	John Paul Recapor	Oct 6, 2026	Oct 26, 2026	◉ Under Review
13	Expected Benefits	Kariel Justine C. Ignacio	Oct 6, 2026	Oct 26, 2026	○ To Do
14	Conclusion and References	Roanne E. Fajardo	Oct 6, 2026	Oct 26, 2026	○ To Do
	Listed in Image 2 only Not in the 14-task list, so it is excluded from the totals				
15	Revision of Project Objectives	Austin Karl C. Garcia	Oct 6, 2026	Oct 20, 2026	? Needs clarification

OCTOBER 2026: T 6 · W 7 · T 8 · F 9 · S 10 · S 11 · M 12 · T 13 · W 14 · T 15 · F 16 · S 17 · S 18 · M 19 · T 20 · W 21 · T 22 · F 23 · S 24 · S 25 · M 26

The schedule above documents the group's actual proposal-writing progress in ClickUp. It is retained as genuine PMIS evidence of the team's documentation work. It is distinct from, and should not be mistaken for, the system-implementation schedule below, which has not yet been configured as a live ClickUp workspace.

Project Implementation Schedule (Proposed)

The chart below translates the Work Breakdown Structure phases (1.1–1.7, shown earlier in this section) into a proposed implementation timeline for system development, aligned with the leads and support members indicated in the WBS diagram. Durations

are expressed in relative project weeks (Week 0 = start of the development phase, following proposal approval) because the instructor has not yet issued an official semester deadline for system development; the team will substitute confirmed calendar dates once available.

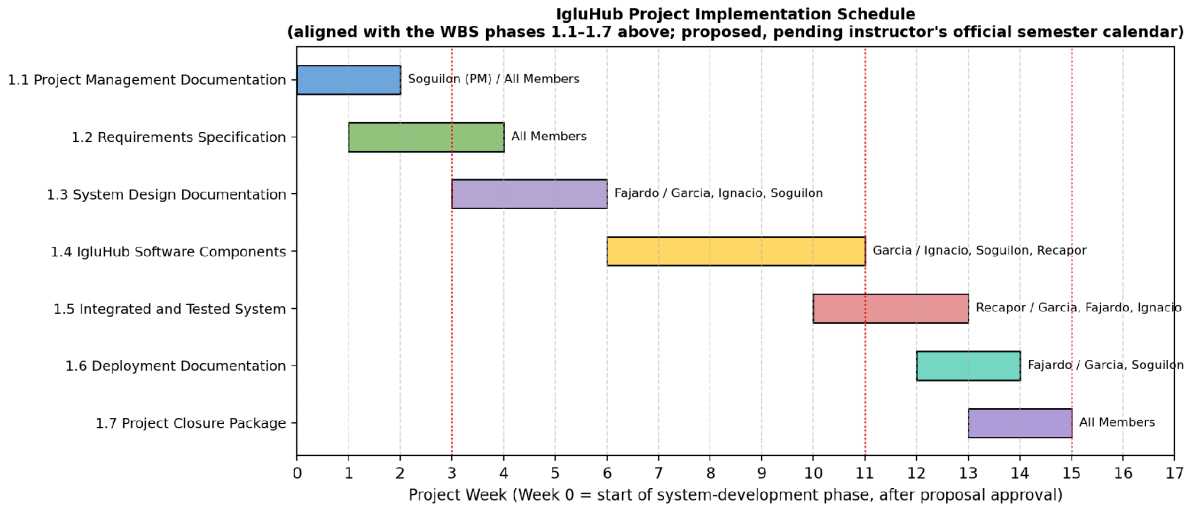


Figure B — Proposed IgluHub System Implementation Schedule, aligned with WBS phases 1.1–1.7.

Resources and Estimated Budget

The development of IgluHub will require human, hardware, software, hosting, internet, testing, and documentation resources for planning, development, testing, and pilot implementation. The estimates below assume that team members will use existing computers and smartphones and will prioritize free development tools and available free-tier services. The amounts are preliminary planning estimates in Philippine pesos (PHP), not vendor quotations, and should be confirmed before expenditure.

A. Required Resources

Resource Category	Description
Human Resources	Project manager, systems analyst, developers, database lead, documentation coordinator, testers, and participating IGLÚ representatives. Team members may perform more than one role.
Hardware Resources	Existing laptops or desktop computers and smartphones for development, browser compatibility checks, and pilot testing.
Software Resources	Visual Studio Code, web browsers, GitHub, ClickUp, and suitable database management/development tools. Free or open-source tools will be prioritized.
Hosting and Database Resources	A web hosting service and centralized cloud database with access controls and backup capabilities. Free tiers will be considered only if they meet the project's functional and security needs.
Internet Resources	Existing internet access and limited mobile data for backup connectivity during development, consultations, and testing.
Testing Resources	Test accounts, synthetic employee records, test products/prices, sample transactions, and compatible devices. Real employee or customer data should not be used in tests unless authorized and appropriately protected.
Documentation Resources	Online document editors, spreadsheets, presentation tools, and limited printing for adviser/client review and testing records.

B. Preliminary Estimated Project Budget

Item	Expense	Basis / Description	Estimated Cost

1	Internet and mobile data	Limited backup data and project-related connectivity	₱300
2	Cloud hosting	Initial deployment and pilot testing; assumes free tier is adequate	₱0
3	Cloud database	Initial centralized storage; assumes an adequate free tier	₱0.
4	Domain name	Optional domain for easier access; may be omitted to reduce costs	₱800
5	Development software	Visual Studio Code, GitHub, browser tools, and other free tools	₱0
6	Testing and printing	Printing questionnaires, review copies, and testing documents	₱500
7	Transportation	Estimated travel for client coordination and pilot testing	₱1,000
8	Miscellaneous expenses	Communication, documentation supplies, and minor unforeseen needs	₱500
	Subtotal		₱3,100
	Contingency fund (10% of subtotal)	Allowance for minor unplanned expenses	₱310
	TOTAL ESTIMATED BUDGET	Preliminary estimate, subject to confirmation	₱3,410

Budget assumptions: The estimate excludes the purchase of computers and smartphones because existing devices will be used. It also excludes paid developer labor and any recurring hosting/database charges beyond the assumed free tier. If free

services prove inadequate, the team must obtain updated quotations and adviser/client approval before incurring additional costs. Actual transportation and printing costs may vary.

C. Budget Control

- The team will record project-related expenses and retain available receipts or other evidence of payment.
- The Project Manager and Technical Planning and Risk Coordinator will review proposed expenses against the approved budget before purchases are made.
- Any expense that exceeds the estimate or changes the project's technical requirements will be discussed with the team and adviser before commitment.
- The contingency fund will be used only for reasonable, project-related unexpected expenses and documented in the final budget report.

Possible Risks and Proposed Solutions

Possible Risks	Proposed Solutions
Inaccurate or incomplete system requirements. The team may misunderstand IGLÚ's actual employee, attendance, cashier, and branch processes, resulting in features that do not match the business needs.	Conduct structured consultations with IGLÚ representatives before development and document confirmed business rules, pricing rules, discount policies, employee assignments, and transaction procedures. Present requirement summaries for validation before proceeding to system design and development.

Scope creep or addition of features beyond the approved project scope. Additional requests, such as payroll, inventory purchasing, accounting, or payment gateway integration, may extend the timeline and exceed the team's capacity.	Maintain the approved scope and limitations document. The Scope and Communication Coordinator will record all requested changes and present them to the team and adviser for review. Features outside the approved scope will be documented as future enhancements rather than immediately developed.
Schedule delays due to simultaneous academic and personal responsibilities. Team members may have limited time for meetings, coding, testing, and documentation, which can delay deliverables.	Use ClickUp to assign tasks, set deadlines, and monitor progress. Divide the work based on each member's designated role, hold regular short progress meetings, and identify delayed tasks early so responsibilities can be adjusted.
Technical difficulties in web and database development. The team may encounter issues in implementing role-based access, centralized data storage, transaction calculations, audit logging, or responsive design.	Conduct early technical planning and use manageable development tools such as Visual Studio Code, GitHub, and suitable database development tools. Develop core features first, test them incrementally, and use version control to track changes and recover from coding errors.

Internet connectivity issues during development, testing, or actual use. Since IgluHub is web-based and relies on a centralized database, poor internet connection may prevent users from accessing the system or saving records.	Inform IGLÚ that a stable internet connection is required for normal system access. During testing, use available internet connections and mobile data as backup. The system documentation will clearly state that offline transaction processing is not included in the current scope.
Data entry errors in employee records, attendance, products, prices, or transactions. Incorrect input may lead to inaccurate reports, wrong totals, or unreliable attendance and sales monitoring.	Implement input validation for required fields, formats, and reasonable values. Provide confirmation screens before submitting transactions and allow authorized personnel to review records. Use test data during development and pilot testing instead of real employee or customer data unless authorized.
Unauthorized access or modification of sensitive records. Employees or other users may attempt to access records beyond their permissions or alter sales and attendance data.	Implement secure login, role-based permissions, and session controls. Restrict sensitive functions such as transaction edits and voids to authorized managers. Maintain an audit log that records the user, action, date and time, affected record, and reason for authorized corrections or voids.

Loss of project or system data due to accidental deletion, system errors, or hosting/database issues. Important development files, source code, or stored records may be lost during the project.	Use GitHub for source-code version control and maintain organized backups of project documents. Use the backup capabilities available from the selected hosting and database provider and document a basic recovery procedure before pilot testing.
Limited budget or unexpected expenses. Costs for printing, transportation, internet data, or optional hosting services may exceed the preliminary estimate of ₱3,410.	Prioritize free development tools and available free-tier hosting or database services. Record all expenses and retain receipts. The Project Manager and Technical Planning and Risk Coordinator will review proposed expenses before purchase, and the contingency fund will be used only for documented project-related needs.
Free hosting or database services may not meet the system's needs. Limited storage, performance restrictions, service downtime, or security limitations may affect pilot testing and system availability.	Evaluate free-tier services based on storage, reliability, security features, backup options, and expected pilot-testing needs before deployment. If the free service is inadequate, prepare updated cost estimates and seek adviser and client approval before spending on paid services.

Low user adoption or difficulty using the system. IGLÚ employees and managers may find the system unfamiliar, leading to incorrect use or resistance during pilot testing.	Design a simple and responsive interface suitable for desktop and mobile browsers. Provide user accounts with permissions appropriate to each role, conduct a brief orientation or demonstration during pilot testing, and gather feedback to identify needed improvements.
Device or browser compatibility issues. The system may not display or function properly on the computers or smartphones used by IGLÚ staff.	Test IgluHub using common desktop and mobile web browsers and the types of devices available to the team and IGLÚ representatives. Address major display or functionality issues before the pilot test and document supported browsers and devices.
Unavailable client representatives or delayed feedback. Delayed responses from IGLÚ may affect requirements validation, testing, and pilot implementation.	Coordinate schedules early through the Scope and Communication Coordinator. Set clear consultation and feedback deadlines, prepare specific questions or demo versions before meetings, and document all confirmed decisions to avoid repeated clarifications.

Insufficient testing before deployment. Undetected errors in attendance recording, sales computation, discounts, payment recording, or permissions may affect system reliability.	Prepare test accounts, synthetic employee records, sample products and prices, and sample transactions. Conduct functional testing for each major feature, including role-based access, calculations, discounts, payment recording, transaction review, and audit logging. Document test results and resolve identified defects before pilot implementation.

Communication Plan

Communication Activity	Roles and Responsibilities	Purpose	Methods of Communication
1. Team Meetings	The Project Manager leads discussions, while all researchers report their progress and concerns.	To coordinate tasks, monitor progress, and solve problems.	Face-to-face meetings or group chat
2. Document Sharing	The Documentation and Schedule Coordinator organizes documents, while all researchers share and review their work.	To keep project documents organized and updated.	Google Drive or shared online documents

3. Adviser Consultation	The researchers present their progress, while the adviser provides guidance and recommendations.	To ensure the project follows the research requirements.	Face-to-face meetings or Online
4. Client Consultation	Researchers communicate with IGLU representatives to understand store operations and confirm the system requirements.	To confirm the system requirements based on IGLÚ's needs.	Interviews, face-to-face meetings, or phone calls
5. System Testing and Feedback	Researchers test IgluHub and gather feedback from selected IGLÚ users to identify errors and areas for improvement.	To identify errors and improve the system's functions.	Hands-on testing and feedback forms

PMIS Tool to Be Used

Project Management Information System (PMIS): ClickUp

ClickUp was selected as the Project Management Information System (PMIS) for the IgluHub project because it provides tools for task assignment, project scheduling, progress monitoring, and team collaboration in one workspace. These functions will

help the five-member project team coordinate responsibilities, organize project activities, monitor deadlines, and manage project documentation throughout the project lifecycle.

Functions and Application to the Project

1. Task Assignment and Management

ClickUp will be used to organize project activities, assign tasks to team members, set deadlines, and monitor task completion. Project activities will be organized according to the Work Breakdown Structure (WBS), allowing the team to identify the tasks required for each project phase. Each task will have an assigned member, a deadline, and a status to clarify individual responsibilities and promote accountability.

2. Project Scheduling and Monitoring

ClickUp provides different project views, including List, Board, Calendar, and Gantt views. These tools will help the team organize tasks, monitor schedules, and track project progress. The Gantt view can be used to visualize task durations and dependencies, while the Board view can help members track activities according to their status. These functions will support the preparation and monitoring of the project's Gantt chart and help the Project Manager identify delayed activities and coordinate schedule adjustments.

3. Team Collaboration and Documentation

ClickUp will serve as a shared workspace where team members can provide progress updates, comment on assigned tasks, and attach relevant files. ClickUp Docs may also be used to organize meeting notes, project requirements, and supporting documentation. These functions will help maintain accessible project information,

reduce confusion regarding task assignments, and improve coordination among members responsible for different project activities.

4. Progress Monitoring and Accountability

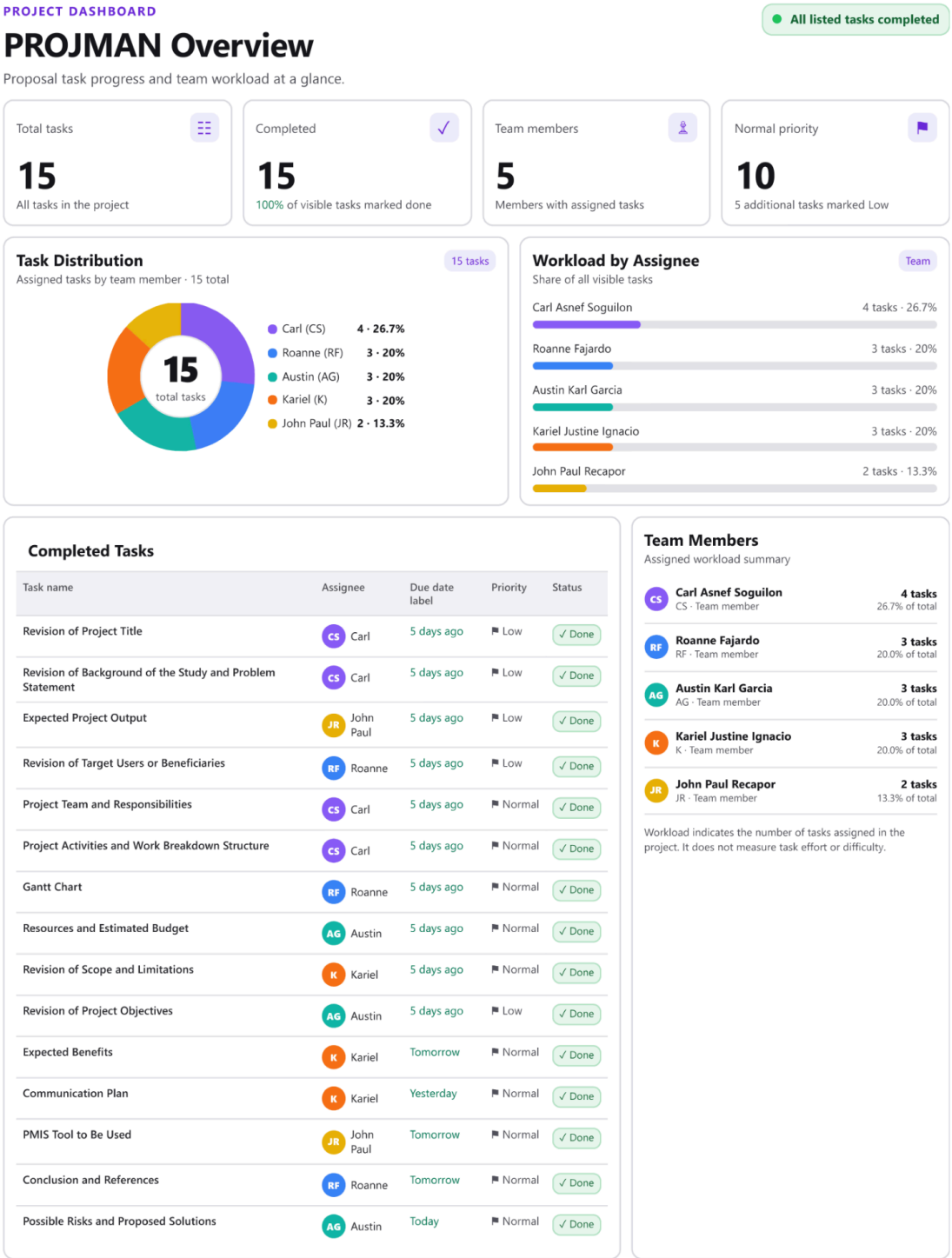
Task statuses, assigned responsibilities, deadlines, and available reporting features will help the team review completed, ongoing, and delayed activities. The Project Manager will use this information to monitor overall progress, follow up on unfinished tasks, and coordinate necessary adjustments. Team members will be responsible for updating their assigned tasks and reporting difficulties that may affect the project schedule.

Justification for Selection

ClickUp is considered suitable for the IgluHub project because its task management, scheduling, and collaboration functions align with the team's project management requirements. Using a shared workspace will help the team organize project records, improve the visibility of task progress, and coordinate documentation and deadlines. These capabilities are particularly relevant to a five-member team with different responsibilities for project planning, documentation, technical analysis, communication, and system output.

However, some ClickUp features may have usage limits or require a paid subscription. The team will prioritize features available under the free plan and use optional paid features only if necessary and financially feasible. The project's essential activities will not depend on paid artificial intelligence features.

Overall, ClickUp will support the planning, coordination, documentation, and monitoring of IgluHub project activities. Its use is expected to help the team maintain organized task records, strengthen accountability, and complete project activities according to the agreed schedule and available resources.


PROJECT DASHBOARD
PROJMAN Overview
Proposal task progress and team workload at a glance.
All listed tasks completed
Total tasks
15
All tasks in the project
Completed
15
100% of visible tasks marked done
Team members
5
Members with assigned tasks
Normal priority
10
5 additional tasks marked Low
Task Distribution
Assigned tasks by team member · 15 total
15 tasks
15
total tasks
Carl (CS) 4 · 26.7%
Roanne (RF) 3 · 20%
Austin (AG) 3 · 20%
Kariel (K) 3 · 20%
John Paul (JR) 2 · 13.3%
Workload by Assignee
Share of all visible tasks
Team
Carl Asnef Soguilon 4 tasks · 26.7%
Roanne Fajardo 3 tasks · 20%
Austin Karl Garcia 3 tasks · 20%
Kariel Justine Ignacio 3 tasks · 20%
John Paul Recapor 2 tasks · 13.3%
Completed Tasks
Task name
Assignee
Due date label
Priority
Status
Revision of Project Title
Carl
5 days ago
Low
Done
Revision of Background of the Study and Problem Statement
Carl
5 days ago
Low
Done
Expected Project Output
John Paul
5 days ago
Low
Done
Revision of Target Users or Beneficiaries
Roanne
5 days ago
Low
Done
Project Team and Responsibilities
Carl
5 days ago
Normal
Done
Project Activities and Work Breakdown Structure
Carl
5 days ago
Normal
Done
Gantt Chart
Roanne
5 days ago
Normal
Done
Resources and Estimated Budget
Austin
5 days ago
Normal
Done
Revision of Scope and Limitations
Kariel
5 days ago
Normal
Done
Revision of Project Objectives
Austin
5 days ago
Low
Done
Expected Benefits
Kariel
Tomorrow
Normal
Done
Communication Plan
Kariel
Yesterday
Normal
Done
PMIS Tool to Be Used
John Paul
Tomorrow
Normal
Done
Conclusion and References
Roanne
Tomorrow
Normal
Done
Possible Risks and Proposed Solutions
Austin
Today
Normal
Done
Team Members
Assigned workload summary
Carl Asnef Soguilon
CS · Team member
4 tasks
26.7% of total
Roanne Fajardo
RF · Team member
3 tasks
20.0% of total
Austin Karl Garcia
AG · Team member
3 tasks
20.0% of total
Kariel Justine Ignacio
K · Team member
3 tasks
20.0% of total
John Paul Recapor
JR · Team member
2 tasks
13.3% of total
Workload indicates the number of tasks assigned in the project. It does not measure task effort or difficulty.

Expected Benefits

The proposed IgluHub system is expected to provide benefits to IGLÚ branch owners, employees, customers, researchers, and future researchers. It aims to improve store operations through organized employee records, attendance monitoring, sales transactions, and centralized branch information.

1. IGLÚ Branch Owners

IgluHub will help branch owners monitor sales, employee attendance, and store operations more easily. It will provide centralized records to support better business decisions.

2. Employees

The system will help employees manage attendance and access employee information more efficiently. It will also assist cashiers in recording sales transactions and processing customer payments.

3. Customers

Customers may experience a smoother transaction process through organized payment calculations and discount applications. This may help reduce errors during checkout.

4. Researchers

The study will help researchers apply their knowledge of information systems in developing a web-based management system. It will also provide practical experience in addressing store management challenges.

5. Future Researchers

This study may serve as a reference for future research on employee monitoring, attendance, and sales management systems. It may also provide ideas for improving similar multi-branch store management systems.

Overall, IgluHub aims to improve store management and provide useful knowledge for future system development.

Conclusion and References

Conclusion

The proposed IgluHub: A Web-Based Multi-Branch Store Management Information System for IGLÚ in Barangay Salapungan, Angeles City aims to address the challenges associated with managing employee records, attendance, cashier transactions, and branch operations through a centralized and organized platform. By integrating these essential business functions into one system, the project seeks to improve the accessibility, consistency, and accuracy of operational information. The proposed system is designed to help authorized personnel manage employee information, monitor attendance, record sales transactions, and access relevant branch records more efficiently, reducing the difficulties associated with maintaining separate records and relying on manual processes.

Furthermore, IgluHub aims to strengthen operational efficiency, data security, and accountability through its centralized database, role-based access controls, audit logs, dashboards, and reporting features. These functions are expected to support more consistent recordkeeping, facilitate the monitoring of sales and employee performance, and provide management with organized information for informed decision-making. By improving coordination between branches and providing a clearer overview of business activities, the system may help management identify operational concerns, review branch performance, and implement appropriate improvements based on available

records. These capabilities are intended to support the daily needs of IGLÚ while establishing more structured and reliable information management practices.

Ultimately, the proposed IgluHub system seeks to provide a practical and sustainable information management solution that can support IGLÚ's current operations and future business development. Its successful implementation will depend on proper system development, comprehensive testing, accurate data entry, reliable internet connectivity, appropriate security measures, and the active cooperation of the project team and business representatives. Through careful implementation and evaluation, IgluHub has the potential to improve the efficiency of multi-branch management, strengthen accountability, and support better coordination across the business. In the long term, the project may serve as a foundation for continuous operational improvement and help prepare IGLÚ for potential business expansion by providing a more organized, consistent, and dependable approach to managing its operations.

References

Aswintama, P., Haryanto, E., & Setyawan, R. A. (2025). Implementation of role-based access control, multi tenancy and audit logging in a single sign-on system. Jurnal Mandiri IT, 14(1), 119–128. https://ejournal.isha.or.id/index.php/Mandiri/article/view/441

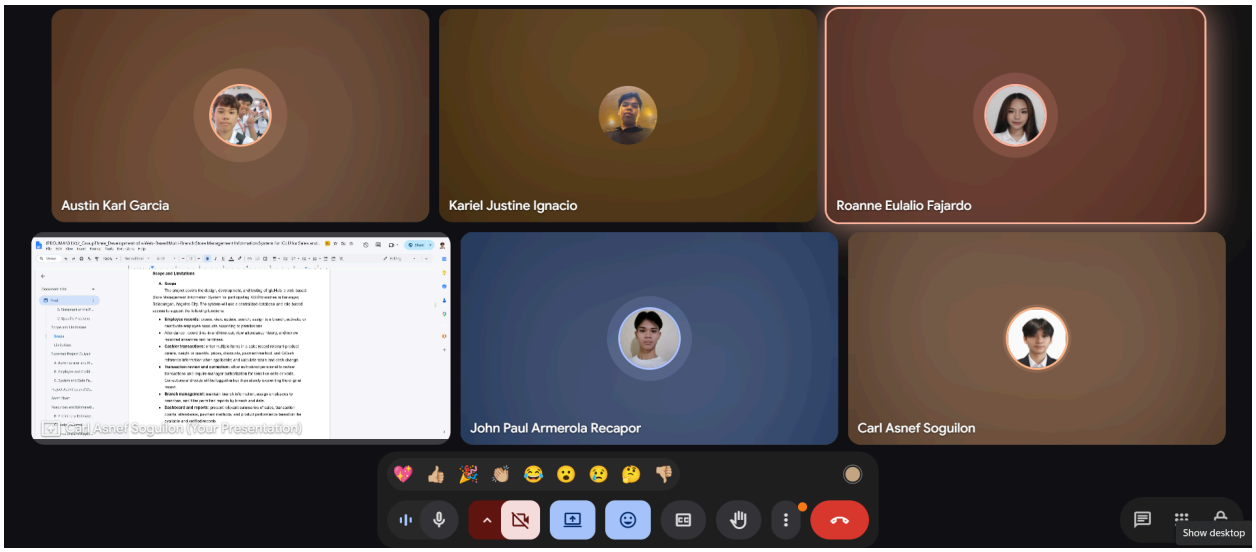
Department of Trade and Industry. (2024, June 26). *DTI pilots "Tindahan Mo, e-Level Up Mo!" to digitalize sari-sari stores in the Philippines.* https://www.dti.gov.ph/dti-news-archived/dti-dti-pilots-tindahan-mo-e-level-up-mo-digitalize-sari-sari-stores-philippines

Mardhatillah, E., Erianda, A., & Hadi, R. (2026). Arsitektur sistem point of sale (POS) multi-cabang berbasis web untuk perusahaan ritel terdistribusi: Studi kasus pada Raffi Collection. *JITSI: Jurnal Ilmiah Teknologi Sistem Informasi, 7*(2), 160–166. https://doi.org/10.62527/jitsi.7.2.606

Oikawa, K., Iwasaki, F., Sawada, Y., & Shinozaki, S. (2025). *Unintended consequences of business digitalization among MSMEs during the COVID-19 pandemic: The case of the Philippines.* Asian Development Bank Institute. https://doi.org/10.56506/BQKT2599

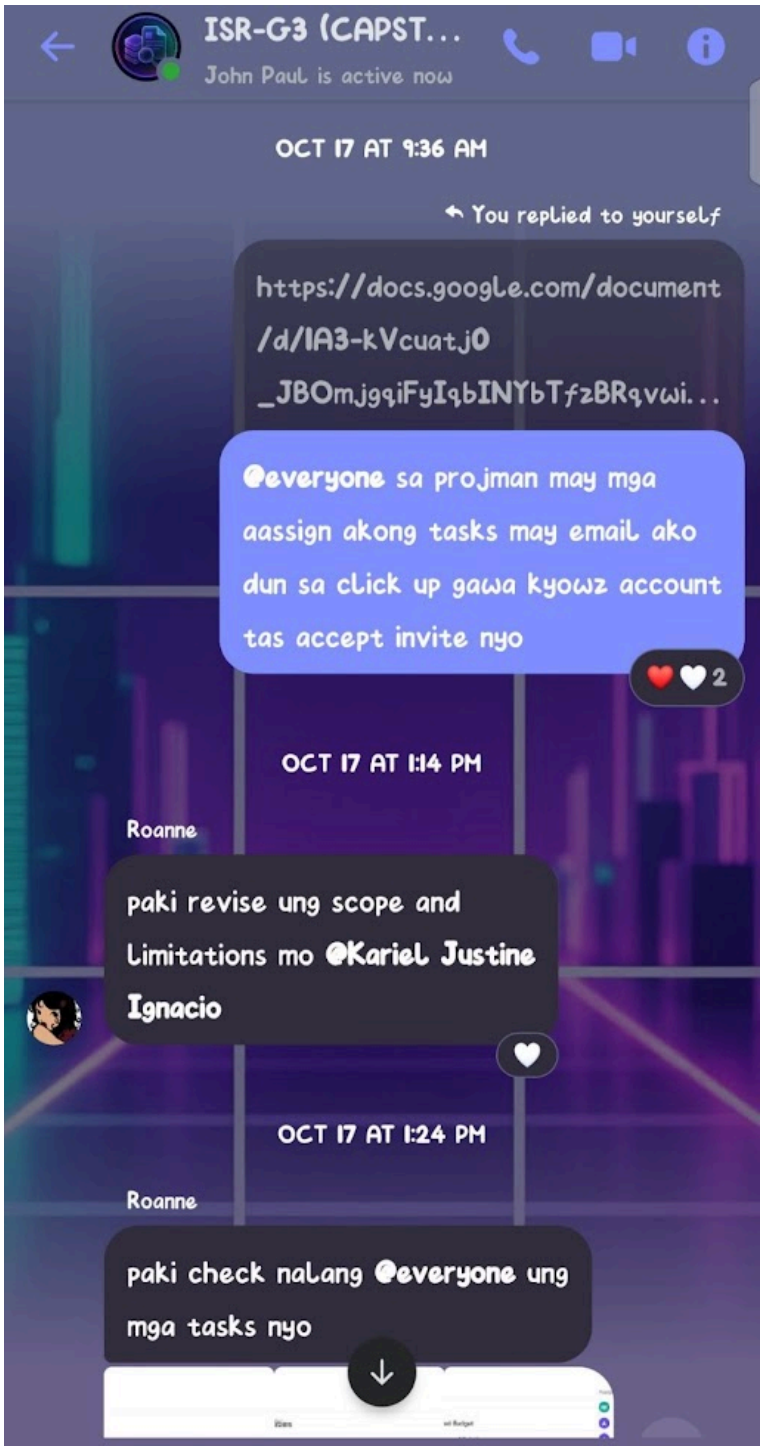
Sagala, G. H., & Őri, D. (2024). Toward SMEs digital transformation success: A systematic literature review. *Information Systems and e-Business Management, 22*(4), 667–719. https://doi.org/10.1007/s10257-024-00682-2

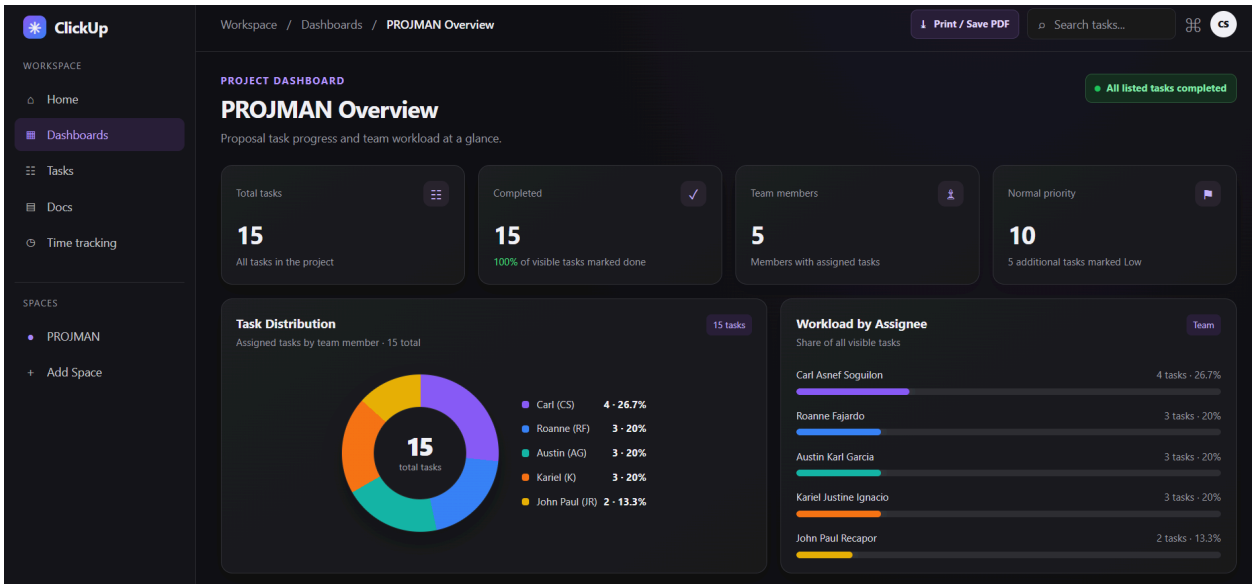
Appendices

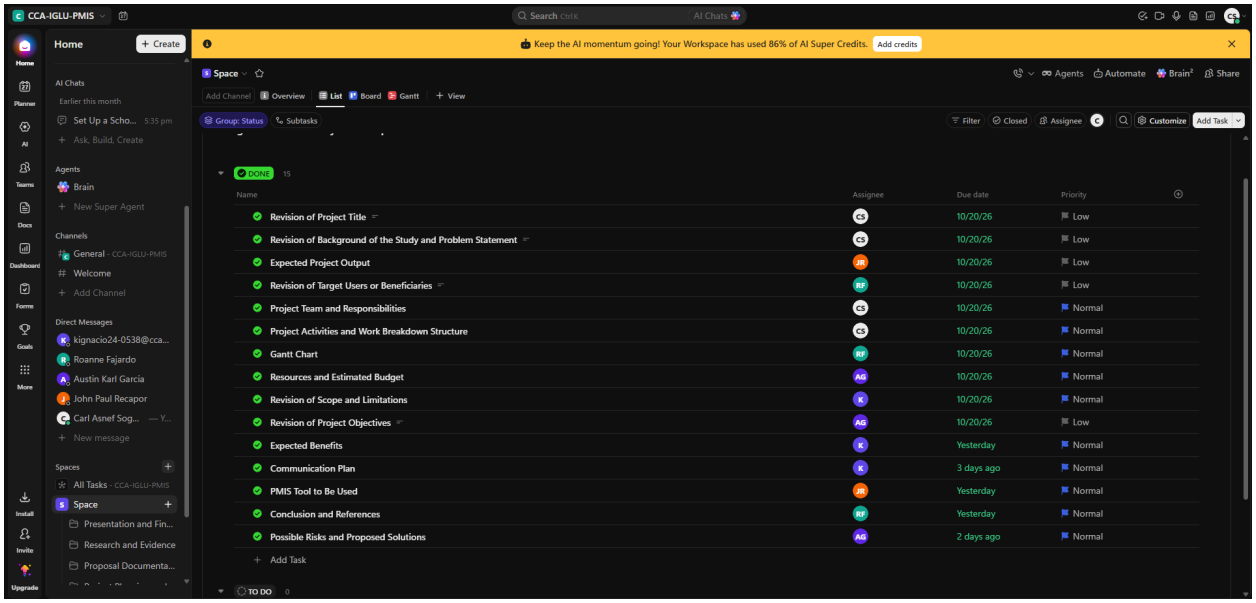

Austin Karl Garcia
Kariel Justine Ignacio
Roanne Eulalio Fajardo
John Paul Armerola Recapor
Carl Asnef Soguilon
Show desktop


Austin Karl Garcia
Kariel Justine Ignacio
Roanne Eulalio Fajardo
John Paul Armerola Recapor
Development of a Web-Based Multi-Branch Store Management Information System for IGLÚ for Sales and Employee Monitoring in Barangay Salapungan, Angeles City
Carl Asnef Soguilon (Your Presentation)
Carl Asnef Soguilon


BE INFORMATION SYSTEMS
ALLIANCE INNOVATION


ISR-G3 (CAPST...
John Paul is active now
OCT 17 AT 9:36 AM
You replied to yourself
https://docs.google.com/document/d/1A3-kVcuatj0_JBOmjgqiFyIqbINYbTfzBRqvwi...
@everyone sa projman may mga aassign akong tasks may email ako dun sa click up gawa kyowz account tas accept invite nyo
2
OCT 17 AT 1:14 PM
Roanne
paki revise ung scope and Limitations mo @Kariel Justine Ignacio
OCT 17 AT 1:24 PM
Roanne
paki check nalang @everyone ung mga tasks nyo


ClickUp
Workspace / Dashboards / PROJMAN Overview
Print / Save PDF
Search tasks...
WORKSPACE
Home
Dashboards
Tasks
Docs
Time tracking
SPACES
PROJMAN
Add Space
PROJECT DASHBOARD
PROJMAN Overview
Proposal task progress and team workload at a glance.
All listed tasks completed
Total tasks
15
All tasks in the project
Completed
15
100% of visible tasks marked done
Team members
5
Members with assigned tasks
Normal priority
10
5 additional tasks marked Low
Task Distribution
Assigned tasks by team member · 15 total
15 tasks
15
total tasks
Carl (CS) 4 · 26.7%
Roanne (RF) 3 · 20%
Austin (AG) 3 · 20%
Kariel (K) 3 · 20%
John Paul (JR) 2 · 13.3%
Workload by Assignee
Share of all visible tasks
Team
Carl Asnef Soguilon 4 tasks · 26.7%
Roanne Fajardo 3 tasks · 20%
Austin Karl Garcia 3 tasks · 20%
Kariel Justine Ignacio 3 tasks · 20%
John Paul Recapor 2 tasks · 13.3%


CCA-IGLU-PMIS
Home
+ Create
Keep the AI momentum going! Your Workspace has used 86% of AI Super Credits.
Add credits
Space
Agents
Automate
Brain²
Share
Overview
List
Board
Gantt
+ View
Group: Status
Subtasks
Filter
Closed
Assignee
Customize
Add Task
DONE 15
Name
Assignee
Due date
Priority
Revision of Project Title 10/20/26 Low
Revision of Background of the Study and Problem Statement 10/20/26 Low
Expected Project Output 10/20/26 Low
Revision of Target Users or Beneficiaries 10/20/26 Low
Project Team and Responsibilities 10/20/26 Normal
Project Activities and Work Breakdown Structure 10/20/26 Normal
Gantt Chart 10/20/26 Normal
Resources and Estimated Budget 10/20/26 Normal
Revision of Scope and Limitations 10/20/26 Normal
Revision of Project Objectives 10/20/26 Low
Expected Benefits Yesterday Normal
Communication Plan 3 days ago Normal
PMIS Tool to Be Used Yesterday Normal
Conclusion and References Yesterday Normal
Possible Risks and Proposed Solutions 2 days ago Normal
Add Task
TO DO 0